

Duolingo’s Practical AI Integration

Generative AI for Business Leaders & C-Suite | Case Study: Build vs Buy in Practice

1. Why Duolingo Matters for Business Leaders

Duolingo is not just a language-learning app — it is a masterclass in strategic AI adoption that every business leader should study. With over eighty million monthly active users and a world-class engineering team, Duolingo had every capability to build AI in-house. They chose not to. Instead, they partnered with OpenAI and integrated GPT-4 into their product in just four months, generating an estimated one hundred eighty million dollars in additional annual revenue.

This case study matters because it demolishes the assumption that sophisticated technology companies must build everything themselves. If Duolingo — a company whose entire product is technology — decided that buying AI was the smarter move, it forces every business leader to ask: why would we build when they did not?

2. The Strategic Question Duolingo Asked

Before making any technology decision, Duolingo's leadership asked a deceptively simple question that every organisation should adopt as a litmus test:

Key Point: "Is building AI our core competitive advantage, or is our advantage in pedagogy, game design, and user engagement?" The answer determined everything that followed. Duolingo's competitive moat is in understanding how people learn languages — not in building AI infrastructure. This distinction is the single most important insight in the entire build-versus-buy debate.

Most organisations struggle with this question because they conflate capability with advantage. Having the capability to build AI does not mean building AI is where your competitive advantage lies. A law firm has lawyers who could write their own document management software, but their advantage is in legal expertise, not software development. The same logic applies to AI.

3. The Product Challenge — Why AI Was Needed

Duolingo's gamified language-learning platform was effective at teaching reading, writing, and listening comprehension. However, it had a fundamental limitation: users could not practise real conversations. Conversational fluency is the most important skill for language learners, and it traditionally requires human tutors — an approach that is impossible to scale to eighty million users across dozens of languages.

The company explored three options:

Option	Estimated Timeline	Estimated Cost	Key Risk
Build custom conversational AI in-house	12–18 months	\$5–\$10 million	High failure risk; diverts engineering from core

			product
Hire human tutors at scale	Ongoing	Billions annually at scale	Economically impossible for 80M users
Partner with an AI provider (OpenAI)	4 months	Under \$2 million first year	Dependency on partner; shared technology

The third option won because it delivered the fastest time to value at a fraction of the cost, with manageable risk.

4. The Partnership Model — How They Did It

In early 2023, Duolingo partnered with OpenAI to integrate GPT-4 into their product. The approach was deliberately pragmatic:

- **No custom model:** They did not train their own large language model. They used OpenAI's existing, proven technology.
- **Wrapper strategy:** Duolingo wrapped GPT-4 in their own pedagogy, user experience, and gamification layer. The AI was the engine; Duolingo's educational expertise was the chassis.
- **Speed over perfection:** The integration went from decision to launch in four months. If they had built custom AI, they would still have been in development eighteen months later.

The resulting product, Duolingo Max, launched with two AI-powered features:

Feature 1 — Explain My Answer

When a user gets an answer wrong, they can ask the AI to explain why. Instead of a generic error message ('Incorrect — the correct answer is...'), the AI provides a personalised, context-aware explanation in natural language. It considers what the user got wrong, explains the grammar rule or vocabulary nuance, and offers examples. This transforms error correction from a frustrating dead-end into an adaptive teaching moment.

Feature 2 — Roleplay Conversations

Users engage in realistic, back-and-forth conversations with AI characters in their target language. The AI adapts to the user's proficiency level, corrects mistakes gently mid-conversation, and keeps the dialogue engaging with contextually appropriate responses. This was the conversational practice that previously required expensive human tutors, now available to every user at scale.

5. The Business Results

Duolingo's AI integration delivered results that exceeded projections across every metric that matters to business leaders.

Metric	Result	Strategic Significance
User engagement	60% increase among Max subscribers	AI features made the product stickier, reducing churn and increasing lifetime value
Premium subscriber growth	35% increase	AI features justified higher pricing; users willingly paid more for obvious value

New revenue stream	~\$15 million/month additional MRR within 6 months	Annualised to ~\$180 million — a material new revenue line from a single feature
Development cost	Under \$2 million first year	Fraction of the \$5–\$10 million custom build estimate
Time to market	4 months from decision to launch	Captured market opportunity while competitors were still planning

The return on investment is stark: under two million dollars invested, one hundred eighty million dollars in new annual revenue generated. Even accounting for ongoing API costs and partnership fees, the margin is extraordinary.

Real-World Incident — Duolingo Max Launch (2023): When Duolingo Max launched, competitors like Babbel and Rosetta Stone were still developing their AI strategies. Duolingo's four-month integration gave them a twelve-to-eighteen-month head start in AI-powered language learning. By the time competitors launched comparable features, Duolingo had already captured market share, refined their AI implementation based on real user data, and established their brand as the AI-forward language platform. First-mover advantage compounded: early data from millions of AI interactions improved the product faster than any competitor could replicate.

6. The Continuous Improvement Dividend

One underappreciated advantage of the partnership approach is what happens after launch. Because Duolingo uses OpenAI's models, every time OpenAI improves GPT — better reasoning, faster responses, lower costs, new languages — Duolingo's product automatically gets better without additional development investment.

If Duolingo had built custom AI, every improvement would require their own engineering resources, testing cycles, and deployment overhead. Instead, their engineers focus entirely on what makes Duolingo unique: pedagogy, gamification, content quality, and user experience. The AI infrastructure improves on someone else's R&D budget.

This is the continuous-improvement dividend: by buying the AI engine, you get vendor R&D for free. Over three to five years, the cumulative value of vendor improvements can be worth more than the initial cost savings.

7. Four Strategic Lessons for Your Organisation

Duolingo's case crystallises four lessons that apply to any industry:

Lesson 1 — Distinguish Core Advantage from Supporting Technology

Duolingo's advantage is in language pedagogy and user engagement — not AI engineering. This distinction drives the buy decision. Ask your leadership team: where do we create unique value that competitors cannot easily replicate? That is where you invest your best talent and resources. Everything else — including AI infrastructure — should be bought, not built.

Lesson 2 — Speed to Market Beats Custom Perfection

Duolingo's four-month launch with partner AI captured a market window that a custom build would have missed entirely. In fast-moving markets, a good-enough solution deployed quickly beats a perfect solution delivered late. First-mover advantage in AI features translates to customer acquisition, brand positioning, and data accumulation that compounds over time.

Lesson 3 — Partnering Provides Continuous Improvement at No Additional Cost

As OpenAI improves GPT, Duolingo's product improves automatically. If they had built custom AI, every improvement would be an engineering project with its own timeline and budget. The partnership model converts a fixed cost (custom development) into a variable benefit (continuous vendor improvement).

Lesson 4 — Focus Your Team on What Differentiates You

Duolingo's engineers work on problems unique to language learning, not on general AI infrastructure that OpenAI does better. This focus creates more value per engineering hour than spreading resources across both the core product and AI infrastructure. The same principle applies to any organisation: every hour your team spends building AI infrastructure is an hour they are not spending on your competitive advantage.

8. Applying the Duolingo Framework to Your Business

Translating Duolingo's approach to your own organisation requires answering four questions:

Question	Duolingo's Answer	Your Organisation's Answer
What is our core competitive advantage?	Language pedagogy, gamification, user engagement	(Identify what you do better than anyone)
Is AI our core product, or does it support our core product?	AI supports the learning product — it is not the product itself	(Most organisations: AI supports, not defines)
Can a commercial AI platform deliver 80%+ of what we need?	Yes — GPT-4 handled conversational AI; Duolingo added pedagogy on top	(Evaluate commercial options honestly)
What is the cost of delay if we build instead of buy?	12–18 months of lost market opportunity and revenue	(Calculate the revenue/efficiency lost during a custom build)

9. Counter-Arguments and When Duolingo's Approach Does Not Apply

Not every organisation should follow Duolingo's exact playbook. The partnership approach is less suitable in situations where:

- **Data sensitivity prevents third-party processing:** A defence contractor or intelligence agency cannot send classified data to OpenAI's API, regardless of data isolation guarantees.
- **Regulatory requirements demand full auditability:** Some industries (healthcare diagnostics, autonomous vehicles) require complete control over AI decision-making for regulatory approval.
- **AI is the product, not a product feature:** If your revenue comes from selling AI capabilities — not from using AI to improve an existing product — building is justified.

- **Vendor dependency creates existential risk:** If OpenAI raised API prices tenfold or discontinued service, Duolingo Max would be crippled. For mission-critical revenue features, this dependency deserves careful risk assessment.

However, these exceptions apply to a small minority of organisations. For the vast majority, the Duolingo model — buy the AI engine, add your domain expertise on top, ship fast — is the optimal strategy.

10. Key Terms Glossary

Term	Definition
Wrapper Strategy	The approach of using a third-party AI model as the underlying engine while adding proprietary domain expertise, user experience, and business logic on top
First-Mover Advantage	The competitive benefit gained by being the first to launch a product or feature in a market, leading to customer acquisition, brand positioning, and data accumulation that compounds over time
Continuous-Improvement Dividend	The ongoing benefit of using commercial AI tools: as the vendor improves their technology, your product improves automatically without additional investment
API (Application Programming Interface)	The technical interface through which Duolingo sends user inputs to OpenAI's GPT-4 and receives AI-generated responses
Monthly Recurring Revenue (MRR)	A subscription-based revenue metric representing the predictable, recurring income generated each month from paying subscribers
Core Competitive Advantage	The specific capability or asset that an organisation does better than competitors and that cannot be easily replicated
Time to Market	The elapsed time between deciding to build or deploy a feature and making it available to customers
Vendor Dependency Risk	The strategic risk that arises when a critical business capability relies on a single external provider whose pricing, availability, or service quality you cannot control

11. Quick Revision Checklist

- Duolingo is a tech company with world-class engineers — and they chose to buy AI rather than build it
- The strategic question: 'Is building AI our competitive advantage, or is our advantage elsewhere?' determines the build-vs-buy decision
- Duolingo partnered with OpenAI and integrated GPT-4 in just 4 months (vs. 12–18 months for custom AI)
- Two AI features launched: Explain My Answer (adaptive error correction) and Roleplay Conversations (scalable conversation practice)
- Results: 60% engagement increase, 35% premium growth, ~\$180 million annualised new revenue, under \$2 million cost
- The wrapper strategy: use commercial AI as the engine, add your domain expertise as the chassis

- Continuous-improvement dividend: vendor R&D improvements flow to you automatically at no additional cost
- First-mover advantage compounds — early data from millions of interactions improves your product faster than competitors can replicate
- Four lessons: distinguish core advantage from supporting tech, speed beats perfection, partnerships provide continuous improvement, focus your team on what differentiates you
- Ask: what is our core advantage? Is AI core or supporting? Can commercial AI deliver 80%+? What is the cost of delay?

20 Exam-Based MCQs — Duolingo's AI Integration

Test yourself after reading. These questions are written in real exam style — scenario-heavy, with plausible distractors. Read every explanation, including for questions you answered correctly.

Q1. What was the strategic question Duolingo asked before deciding whether to build or buy AI?

- A) Can we afford to hire enough AI engineers?
- B) Is building AI our core competitive advantage, or is our advantage in pedagogy and engagement?
- C) Will investors prefer a custom AI solution over a partnership?
- D) Can OpenAI's technology handle multiple languages?

Answer: B **Explanation:** B is correct — Duolingo framed the decision around core competitive advantage, not cost or capability. They could afford to build and had the talent, but recognised their advantage was in understanding how people learn, not in AI infrastructure. A focuses on cost, which was secondary. C focuses on investor optics. D is a technical question, not a strategic one.

Q2. *Scenario:* A fintech startup has built a successful budgeting app with 5 million users. The product team wants to add an AI financial advisor feature. The CTO advocates building custom AI, arguing that financial advice requires specialised models. The CEO points out that GPT-4 already handles financial reasoning well and suggests a partnership approach similar to Duolingo's. Based on the Duolingo framework, which approach is MOST appropriate?

- A) Build custom AI because financial data is too sensitive for third-party processing
- B) Partner with an AI provider and wrap the AI in the startup's financial expertise and UX, similar to Duolingo wrapping GPT-4 in pedagogy
- C) Abandon the AI advisor feature entirely because fintech is too regulated for AI
- D) Wait for a fintech-specific AI platform to emerge before adding the feature

Answer: B **Explanation:** B is correct because the startup's competitive advantage is in financial UX and user trust, not in AI infrastructure — directly paralleling Duolingo's situation. The wrapper strategy (commercial AI engine + proprietary domain expertise) maximises speed to market while preserving focus. A assumes data sensitivity prevents all third-party use, but enterprise AI providers offer SOC 2 compliance and data isolation. C is overly cautious. D cedes first-mover advantage.

Q3. How long did it take Duolingo to go from the partnership decision to launching Duolingo Max?

- A) Two months
- B) Four months
- C) Twelve months
- D) Eighteen months

Answer: B **Explanation:** B is correct — the integration took four months from decision to launch. This contrasts sharply with the estimated twelve to eighteen months for custom AI development. A is too fast for enterprise-grade integration. C and D represent the timeline that would have been required for a custom build.

Q4. *Scenario:* A healthcare company's board is debating AI strategy. The Chief Medical Officer argues they should build custom AI for patient engagement because 'our clinical protocols are unique.' The Chief Digital Officer counters that several patient-engagement AI platforms already exist with HIPAA compliance and could be customised to their protocols. What Duolingo lesson is MOST relevant to this debate?

- A) First-mover advantage — they should launch the fastest option
- B) Distinguish core advantage from supporting technology — if their advantage is clinical expertise, they should buy the AI engine and add their protocols on top
- C) The continuous-improvement dividend — vendor AI will improve faster than custom AI
- D) Speed to market — four months is better than eighteen months

Answer: B **Explanation:** B is correct because the fundamental question is whether AI engineering is the hospital's competitive advantage (it is not — clinical expertise is). Just as Duolingo wrapped GPT-4 in pedagogy, the healthcare company should wrap commercial AI in their clinical protocols. A, C, and D are valid points but secondary to the core advantage question, which must be answered first.

Q5. What is the 'continuous-improvement dividend' in the context of AI partnerships?

- A) The interest earned on money saved by not building custom AI
- B) The ongoing benefit that your AI capabilities improve automatically as the vendor improves their technology, without additional investment from you
- C) The annual price reduction that AI vendors contractually guarantee
- D) The improvement in employee skills that comes from using AI tools daily

Answer: B **Explanation:** B is correct — when OpenAI improves GPT, Duolingo's product improves automatically at no additional cost. This is the continuous-improvement dividend: vendor R&D benefits flow to all customers. A misinterprets the term financially. C is not guaranteed contractually. D describes employee development, not the vendor-improvement dynamic.

Q6. *Scenario:* A media company launches an AI-powered content recommendation engine using a commercial AI platform. Six months later, the vendor releases a major model upgrade that improves

recommendation accuracy by 25%. The media company's product benefits immediately without any engineering work. What concept does this illustrate?

- A) Vendor lock-in
- B) The continuous-improvement dividend
- C) First-mover advantage
- D) The wrapper strategy

Answer: B **Explanation:** B is correct — the vendor's R&D investment directly improves the customer's product without requiring the customer's own engineering resources. This is the continuous-improvement dividend in action. A is the downside of vendor dependency, not what is illustrated here. C refers to being first to market. D refers to the approach of layering proprietary expertise on top of commercial AI.

Q7. What was Duolingo's estimated first-year cost for the OpenAI partnership?

- A) Under \$500,000
- B) Under \$2 million
- C) \$5–\$10 million
- D) \$15–\$20 million

Answer: B **Explanation:** B is correct — Duolingo's total first-year cost including API fees and partnership fees was under \$2 million. C (\$5–\$10 million) is what a custom build was estimated to cost. A is below the actual figure. D is far above the actual cost.

Q8. *Scenario:* *An e-commerce company's CTO proposes building a custom AI product recommendation system. He estimates it will take 14 months and cost \$4 million, but will deliver 'best-in-class' recommendations. Meanwhile, a competitor has just launched AI-powered recommendations using a commercial platform and is already seeing a 20% increase in average order value. What Duolingo lesson does this scenario illustrate?*

- A) Focus your team on what differentiates you
- B) Speed to market beats custom perfection — a good-enough solution deployed quickly captures market opportunity that a perfect solution delivered late will miss
- C) The continuous-improvement dividend
- D) Core advantage should never involve AI

Answer: B **Explanation:** B is correct — the competitor's commercial solution is already generating revenue and data while the custom build has not started. Duolingo's four-month launch with partner AI beat the eighteen-month timeline for custom AI, capturing market opportunity and revenue. A is a valid lesson but not what this scenario primarily illustrates. C is about vendor R&D benefits. D is an overstatement.

Q9. What strategy did Duolingo use to integrate OpenAI's technology?

- A) They trained their own large language model using OpenAI's tools
- B) They replaced their entire platform with OpenAI's technology

- C) They used GPT-4 as the AI engine and wrapped it in their own pedagogy, gamification, and user experience
- D) They hired OpenAI engineers to build a custom solution exclusively for Duolingo

Answer: C **Explanation:** C is correct — the wrapper strategy: GPT-4 provides the conversational AI capability, Duolingo adds the pedagogical structure, gamification, content quality, and user experience that make it effective for language learning. A incorrectly describes building a model. B overstates the integration scope. D mischaracterises the partnership.

Q10. Scenario: *A consulting firm's managing partner is concerned about using commercial AI because 'if OpenAI raises prices significantly, we will be dependent on them.' She wants to build in-house AI instead. How should this vendor dependency risk be BEST addressed?*

- A) Build all AI in-house to eliminate any vendor dependency
- B) Acknowledge the risk but weigh it against the much larger risk of spending millions on custom AI that has a 70–80% failure rate — and mitigate vendor risk through contract terms, multi-vendor strategy, and monitoring alternatives
- C) Ignore the risk because AI vendors will never raise prices
- D) Switch to a different AI vendor preemptively before prices increase

Answer: B **Explanation:** B is correct because vendor dependency is a real risk but should be managed proportionally, not avoided by taking on the much larger risk of custom AI development failure. Mitigation strategies include negotiating multi-year contracts, maintaining awareness of alternative vendors, and structuring integration code to be vendor-switchable. A overreacts to a manageable risk. C is naive. D is premature without evidence of impending price increases.

Q11. What was the approximate annualised additional revenue generated by Duolingo Max within six months of launch?

- A) \$10 million
- B) \$50 million
- C) \$180 million
- D) \$500 million

Answer: C **Explanation:** C is correct — approximately \$15 million in additional monthly recurring revenue within six months, annualised to approximately \$180 million. A and B significantly understate the figure. D overstates it.

Q12. Scenario: *A university's IT department is debating whether to build or buy AI-powered tutoring for their online courses. The Dean says: 'We have a computer science department with AI researchers — we should build it ourselves.' The Provost responds: 'Our competitive advantage is in our faculty's teaching expertise and our curriculum, not in AI engineering.' Whose reasoning better aligns with the Duolingo framework?*

- A) The Dean — having internal AI expertise means they should always build

- B) The Provost — the university's advantage is in pedagogy and curriculum, not AI infrastructure, so they should buy the engine and add their expertise on top
- C) Neither — universities should not use AI for tutoring
- D) Both — they should build AI using their researchers and also buy commercial tools simultaneously

Answer: B **Explanation:** B is correct because it directly parallels Duolingo's reasoning: having the capability to build does not mean building is where your advantage lies. Duolingo had world-class engineers and still chose to buy. The university's advantage is in teaching expertise and curriculum — they should wrap commercial AI in that expertise. A confuses capability with advantage. C is an irrelevant prohibition. D wastes resources on parallel tracks.

Q13. Which AI-powered feature does Duolingo Max's 'Explain My Answer' replace?

- A) Human tutor explanations delivered via video call
- B) Generic error messages that tell users the correct answer without explaining why
- C) Pre-written grammar explanations from a textbook database
- D) Peer-to-peer learning forums where users explain answers to each other

Answer: B **Explanation:** B is correct — before Explain My Answer, users who answered incorrectly received generic error messages ('Incorrect — the correct answer is...'). The AI feature replaces these with personalised, context-aware explanations that consider what the user got wrong and why. A describes a service Duolingo never offered at scale. C is static content, not what was replaced. D describes community features, not error handling.

Q14. *Scenario:* A SaaS company's VP of Engineering argues against using commercial AI, saying: 'Our competitors will have the same AI capabilities if we all use the same vendor. We need custom AI for differentiation.' The CEO responds: 'Duolingo used the same GPT-4 that anyone can access, yet they differentiated through how they applied it.' Who is correct, and why?

- A) The VP — using the same AI vendor as competitors provides no differentiation
- B) The CEO — differentiation comes from how you apply AI to your domain expertise, not from the underlying AI engine itself
- C) Neither — differentiation in AI is impossible because all models produce identical outputs
- D) The VP — only custom AI can create lasting competitive advantage

Answer: B **Explanation:** B is correct because Duolingo proved that differentiation comes from the wrapper (pedagogy, UX, gamification), not the engine (GPT-4). Any competitor could access GPT-4, but none could replicate Duolingo's fifteen years of language-learning expertise, user data, and engagement mechanics. A ignores the wrapper-strategy evidence. C is technically wrong — application context changes output quality dramatically. D overstates custom AI's exclusivity.

Q15. What percentage increase in user engagement did Duolingo Max subscribers show?

- A) 15%

- B) 35%
- C) 60%
- D) 90%

Answer: C **Explanation:** C is correct — user engagement increased 60% among Duolingo Max subscribers. B (35%) is the premium subscriber growth figure, not the engagement increase. A and D are incorrect.

Q16. Scenario: A logistics company decides to follow the Duolingo model. They identify route optimisation as a key area for AI. Instead of building custom AI, they subscribe to a commercial route-optimisation platform and add their proprietary delivery constraints (time windows, vehicle capacities, driver preferences) on top. Which Duolingo concept are they applying?

- A) First-mover advantage
- B) The wrapper strategy — using commercial AI as the engine and adding domain-specific expertise on top
- C) The continuous-improvement dividend
- D) The lean AI team model

Answer: B **Explanation:** B is correct — the logistics company is using commercial AI as the engine (route optimisation algorithms) and wrapping it in their domain-specific constraints (delivery windows, vehicle specs, driver preferences). This directly mirrors Duolingo wrapping GPT-4 in language-learning pedagogy. A, C, and D are valid concepts but do not describe this specific approach.

Q17. What is the PRIMARY risk of the partnership approach that Duolingo accepted?

- A) Users would prefer a competitor's AI features
- B) Dependency on OpenAI — if OpenAI raised prices significantly, discontinued service, or degraded quality, Duolingo Max would be affected
- C) OpenAI would steal Duolingo's pedagogy and create a competing product
- D) Regulatory agencies would prohibit AI-powered language learning

Answer: B **Explanation:** B is correct — vendor dependency is the acknowledged risk of the partnership model. If OpenAI's pricing, availability, or quality changed unfavourably, Duolingo Max would be directly impacted. This risk is managed but not eliminated. A is a competitive risk unrelated to the partnership model. C is mitigated by contractual protections. D is not a realistic regulatory concern for language learning.

Q18. Scenario: A retail chain's CEO says: 'Duolingo is a tech company — their example does not apply to us. We are a brick-and-mortar retailer.' The CIO responds: 'That is exactly why we should follow their model. If a tech company with world-class engineers chose to buy AI, a retailer with no AI engineering capability should absolutely buy rather than build.' Is the CIO's logic sound?

- A) No — retail and technology are too different for the comparison to be valid

- B) Yes — if Duolingo, a company with the capability to build, chose to buy because their advantage was elsewhere, a retailer whose advantage is even further from AI engineering has an even stronger case for buying
- C) No — retailers should not use AI at all because their advantage is in physical store experience
- D) Yes — but only because retailers cannot afford custom AI development

Answer: B **Explanation:** B is correct because the CIO's argument uses Duolingo as an upper bound: if even a technology company chose not to build AI, the case for buying is even stronger for non-technology companies whose competitive advantage is further from AI engineering. A dismisses a valid analogy without justification. C is overly restrictive. D is partially true but misses the strategic argument — it is about advantage, not just affordability.

Q19. What problem did Duolingo's Roleplay Conversations feature solve that was previously impossible to address at scale?

- A) Teaching grammar rules to non-native speakers
- B) Providing real conversational practice that traditionally required human tutors, now available to 80 million users in dozens of languages
- C) Translating documents between languages
- D) Grading written essays in foreign languages

Answer: B **Explanation:** B is correct — conversational fluency practice was Duolingo's biggest product gap because it required human tutors, which could not scale to 80 million users across dozens of languages. AI made this possible at negligible marginal cost per user. A, C, and D are language-related tasks but were not the specific problem the Roleplay feature addressed.

Q20. *Scenario:* A board member reviews Duolingo's AI case study and asks: 'What would have happened if Duolingo had chosen to build custom AI instead of partnering with OpenAI?' Based on the case study, which outcome was MOST likely?

- A) They would have built a superior AI system within 6 months
- B) They would still be in development 12–18 months later, having spent \$5–\$10 million, while competitors who chose partnerships captured the market opportunity
- C) The custom AI would have cost less than the partnership in the long run
- D) The custom AI would have worked but produced identical results to the partnership approach

Answer: B **Explanation:** B is correct because the estimated custom build timeline was 12–18 months at a cost of \$5–\$10 million — with no guarantee of success given the 70–80% failure rate for custom AI projects. During that time, competitors using commercial AI would have captured market share. A is unrealistic given typical timelines. C ignores the ongoing maintenance costs of custom AI. D ignores the failure risk and the opportunity cost of delayed launch.